

CA Note

1 Basic Knowledge

Group $(G, *)$: **G1**: Closure **G2**: Associativity **G3**: Identity element **G4**: Inverse element $\downarrow$ **Division ring**: $R^\times = R \setminus \{0\}$; $R^\times$ group (under $\cdot$)

Ring $(R, +, \cdot)$: **R1**: $(R, +)$ is an abelian group **R2**: $(R, \cdot)$ is monoid. (no inverse) **R3**: Distributive law (left and right)

Ideal (I) : **I1**: $\neq \emptyset$ **I2**: $\forall a, b \in I, a - b \in I$ **I3**: $\forall r \in R, i \in I, ri, ir \in I$ **Lemma**: $I = R \Leftrightarrow 1 \in I \Leftrightarrow I \cap R^\times \neq \emptyset$

Subgroup (H) : **H1**: $e \in H$ **H2**: $\forall a, b \in H, ab \in H$ **H3**: $\forall a \in H, a^{-1} \in H$ **Subring** (S) : **S1**: $1 \in S$ **S2**: $\forall a, b \in S, a - b \in S$ **S3**: $\forall a, b \in S, ab \in S$

Integral Domain **ID**: **ID1**: commutative ring **ID2**: no zero divisors **Lemma**: Finite ID $\Rightarrow$ field ID $\Rightarrow$ Cancellation Law

Factor Ring (R/I) : $a + I = b + I \Leftrightarrow a - b \in I$ $(R/I, \oplus, \odot)$ is a ring. ($1_{R/I} = 1_R + I, 0_{R/I} = 0_R + I, I$ ideal) $\mathbb{Z}_n \cong \mathbb{Z}/n\mathbb{Z}$

Ring Homomorphism: $\varphi: R \rightarrow S$ s.t. **RH1**: $\varphi(a + b) = \varphi(a) + \varphi(b)$ **RH2**: $\varphi(ab) = \varphi(a)\varphi(b)$ **RH3**: $\varphi(1_R) = 1_S$

$\cdot$ **1**: $\text{Ker}(\varphi)$ is an ideal. **2**: $\text{Im}(\varphi)$ is a subring. **3**: φ is 1-1 $\Leftrightarrow \text{Ker}(\varphi) = \{0_R\}$. **FIT**: $R/\text{Ker}(\varphi) \cong \text{Im}(\varphi)$ (First Isomorphism Theorem)

$\cdot$ **Canonical Homomorphism**: $\text{can}: R \rightarrow R/I$ s.t. $\text{can}(r) = r + I$ is a onto ring Homomorphism. $\text{Ker}(\text{can}) = I \quad \updownarrow (\exists! \tilde{\varphi} \text{ 双射})$

$\cdot$ **Lemma**: If $\varphi: R \rightarrow S$ ring Homomorphism s.t. $I \subseteq \text{Ker}(\varphi)$, then $\exists! \tilde{\varphi}: R/I \rightarrow S$ s.t. $\varphi = \tilde{\varphi} \circ \text{can}$ (i.e. $\tilde{\varphi}(r + I) = \varphi(r)$)

Lagrange's Theorem: G finite group, $H \leq G$ subgroup, then $|H| \mid |G|$

2 Types of Ideals

Division: If R ring, then: $\forall f, g \in R[x], g$ 首相系数可逆 $\Rightarrow \exists q, r \in R[x]$ s.t. $f(x) = g(x)q(x) + r(x)$ and $\deg(r) < \deg(g)$ or $r = 0$.

Lemma: If R ring, $g \in R[x], g \neq 0, I = (g)$. $\Rightarrow \overset{1}{\forall i \in R[x]/I, i = f + I, \deg(f) < \deg(g)}$

g 首相系数可逆 $\overset{2}{\text{If } a, b \in R[x], \deg(a, b) < \deg(g) \text{ and } a + I = b + I, \text{ then } a = b. (\text{系数也相同})}$

Remark: If $I = (g), g$ 首相系数不可逆: Find a polynomial g_1 s.t. $g_1 = hg, h \in R[x]$ 首相系数可逆, 那么可满足 $\overset{1}{\text{条件}}$, $\overset{2}{\text{只能 subset 但不能唯一表达}}$.

Remark: 有些时候我们能用 gcd, lcm 计算 $\langle a, b \rangle = \langle \gcd(a, b) \rangle$ 和 $a \cap b = \langle \text{lcm}(a, b) \rangle$ [If R commutative ring] 但注意! 要在能够用 Division Algorithm | 有 gcd, lcm 的环里用.

Principal Ideal: Ideal I : Iff $I = aR$ **Prime Ideal**: Ideal $I \neq R$: $ab \in I \Rightarrow a \in I$ or $b \in I$. (OR: If $a \notin I$ and $b \notin I$, then $ab \notin I$)

Maximal Ideal: Ideal $I \neq R$: If J ideal and $I \subseteq J \subseteq R$, then $J = I$ or $J = R$. $\downarrow I$ not radical $\Leftrightarrow \exists a \notin I$ s.t. $a^n \in I$ for some $n \in \mathbb{N}$

Radical|**Radical Ideal**: I : **Radical**: $\sqrt{I} = \{r \in R : r^n \in I \text{ for some } n \in \mathbb{N}\}$ **Radical Ideal**: If $I = \sqrt{I}$ $I \subseteq \sqrt{I}$ always.

Properties: Relationship between these kinds of ideals: **Maximal Ideals** $\subseteq$ **Prime Ideals** $\subseteq$ **Radical Ideals** $\subseteq$ **Ideals**

1. R commutative ring, I ideal. **I prime $\Leftrightarrow R/I$ is an ID.** **I maximal $\Leftrightarrow R/I$ is a field** **I maximal $\Rightarrow I$ prime**

2. R ring, $I_1, I_2, \dots, I_n \triangleleft R, I_1 \cap I_2 \cap \dots \cap I_n = \{0\} \Rightarrow R$ isomorphic to a subring of $R/I_1 \oplus R/I_2 \oplus \dots \oplus R/I_n$. **$R \cong R/I_1 \oplus R/I_2 \oplus \dots \oplus R/I_n$** if $I_i + I_j = R$ for $i \neq j$

3. R ring, I ideal, J ideal of $R/I, T = \{r \in R : r + I \in J\}$ (reps of J). Then $\overset{1}{T: \text{Ideal of } R}$ $\overset{2}{I \subseteq T}$ $\overset{3}{J = \{t + I : t \in T\}}$ $\overset{4}{(R/I)/J \cong R/T}$
If $J = \langle j_1 + I, \dots, j_n + I \rangle$ then $T = \langle j_1, \dots, j_n \rangle + I = \langle j_1, \dots, j_n, I \rangle$ Moreover, if $I = \langle i_1, \dots, i_m \rangle$, then $T = \langle j_1, \dots, j_n, i_1, \dots, i_m \rangle$

4. R commutative ring, I ideal. **I radical $\Leftrightarrow R/I$ has no nilpotent elements** except 0 **I prime $\Rightarrow I$ radical** (nilpotent: $r^n = 0$)

5. If R (commutative with unity), then any ideal $I \neq R$ is contained in a maximal ideal. If r is not a unit, then $1 \notin \langle r \rangle = rR$

3 PID | UFD | Euclidean Domain (ED $\Rightarrow$ PID $\Rightarrow$ UFD $\Rightarrow$ ID)

Irreducible: $0 \neq a \in R \setminus R^\times$ is **Irreducible**: If $a = bc$, then $b \in R^\times$ or $c \in R^\times$.

Prime: $0 \neq a \in R \setminus R^\times$ is **Prime**: If $a \mid bc$, then $a \mid b$ or $a \mid c$. (If R commutative ring, then prime $\Rightarrow$ irreducible)

1. F field, $0 \neq g \in F[x]$ non-constant: $\overset{1}{F[x]/\langle g \rangle \text{ is a field}} \Leftrightarrow \overset{2}{g \text{ is irreducible in } F[x]} \Leftrightarrow \overset{3}{F[x]/\langle g \rangle \text{ is a domain}}$

2. F field, $0 \neq g \in F[x]$: **g is irreducible $\Leftrightarrow \langle g \rangle$ is prime ideal**

3. If R is ID: **Prime $\Rightarrow$ Irreducible** If $0 \neq a \in R$, then: **a prime $\Leftrightarrow aR$ prime ideal**

4. If R is PID, $a \neq 0$: $\overset{1}{aR \text{ is prime ideal}} \Leftrightarrow \overset{2}{a \text{ is prime}} \Leftrightarrow \overset{3}{aR \text{ is maximal ideal}} \Rightarrow \overset{4}{a \text{ is irreducible}}$

PID: R commutative ring is principal ideal domain if: $\forall I \triangleleft R, I = aR$ for some $a \in R$. Theorem: $F, F[x]$ field is PID.

ED: R is ID. R is Euclidean Domain if: (**Euclidean Function**) $\exists v: R \setminus \{0\} \rightarrow \mathbb{N}_0$ s.t. $\forall a, b \in R, b \neq 0, \exists q, r \in R$ s.t. $a = bq + r$ and $r = 0$ or $v(r) < v(b)$.

Common ED: $\mathbb{Z}, F[x], \mathbb{Z}[i]$ Gaussian Integers are ED, with $v(a) = |a|, v(f) = \deg(f), v(a + bi) = a^2 + b^2$ respectively.

Euclidean Algorithm: If R ED, $a, b \in R \setminus \{0\}$, then $\gcd(a, b) = xa + yb$ for some $x, y \in R$. (extended of Bezout's Identity)

Bezout's Identity: R is PID, then $\forall a, b \in R, \exists x, y \in R$ s.t. $ax + by = \gcd(a, b)$.